

Table 1: The functionalities of Cloud Foundry, Cloudify and OpenShift components

Functionality	Cloud Foundry	Cloudify	OpenShift
Core functionality	Cloud controller	Manager	Broker
Providing third party database services	Service broker	Agent	Cartridge
Routing of incoming traffic	Router	Manager	REST API
Querying the state of apps	Cloud controller	CLI client	Broker
Messaging	Message bus	Manager	Broker
App instance management	Droplet execution agent	Agent	Node
Application state management	Health manager	Manager	Broker
Containerisation	Warden	Agent	Gear
Load balancing of user requests	Droplet execution agent	Manager	Broker
Framework provider	Blob store	Agent	Cartridge
Tech languages	Java, Ruby, Scala, Node.js, Groovy, Grails, PHP, Go, Python	Java, PHP, Ruby	Java, Ruby, Node.js, PHP, Python, Perl, JavaScript
Databases	MongoDB, MySQL, PostgreSQL	MySQL, MongoDB	MongoDB, MySQL, PostgreSQL
Frameworks	Spring, Rails, Grails, Play Sinatra	JavaScript, Node.js	Rails, Flask, Django, Drupal, Vertx
Horizontal scaling	Yes	Yes	Yes
Vertical scaling	Yes	No	Yes
Auto scaling	Yes	Yes	Yes

c) **Cloudify:** This was developed and designed on the principles of openness, in order to power the IT transformation revolution. It enables organisations to design, build and deliver various business applications and network services. The latest version of Cloudify is 4.3, which incorporates enhanced features like advanced security, control and true self-service. Cloudify 4.3 has introduced a totally new concept for container orchestration with Kubernetes (Kubernetes Cloud Native Orchestration).

Table 1 shows the basic functionalities of the Cloud Foundry, Cloudify and OpenShift architectural components. This is purely based on our views and the authenticity of the features should ideally be revalidated by the cloud provider.

To summarise, PaaS is a computing platform that enables the quick and easy creation of Web applications without the necessity of buying and maintaining the underlying software and infrastructure.

From the industry adoption statistics (as shown earlier in the article), it can be seen that PaaS adoption is picking up very rapidly. PaaS provides a huge relief to enterprises by helping them bring down their capital and operational expenditures when delivering their services. It enables the enterprise applications to be run on any cloud platform, whether public or private. This means that a PaaS application developed on Amazon AWS can easily be ported to Microsoft Azure, and from there to VMware vSphere and to Red Hat RHEV, etc.

PaaS is useful when multiple developers are working on a project or when other external users need to collaborate

with the development process, i.e., it is best suited for agile software development, because it eases the difficulties around the rapid development of software. **END** 

References

- [1] https://i2.wp.com/www.vamsitalkstech.com/wp-content/uploads/2016/10/451-Research-PaaS_vs_SaaS_IaaS.jpg
- [2] <https://www.gartner.com/newsroom/id/3871416>
- [3] <https://www.forbes.com/sites/louiscolombus/2018/01/07/83-of-enterprise-workloads-will-be-in-the-cloud-by-2020/#2ffc2cd66261>
- [4] IDC Future Scape: Worldwide IT Industry 2018 Predictions -US43171317.pdf

Acknowledgements

The authors would like to thank Hari Kishan Burle and Raju Alluri of the Global Enterprise Architecture Group of Wipro Technologies for giving their time and support to bring out this article.

By: Dr Gopala Krishna Behara and Tirumala Khandrika

Dr Gopala Krishna Behara is a lead enterprise architect in the Global Enterprise Architecture division of Wipro. He has 21 years of experience in the IT field. He can be reached at gopalkrishna.behara@wipro.com.

Tirumala Khandrika is a senior architect in the SCA Practice division of Wipro with 16 years of IT experience. He can be reached at tirumala.khandrika@wipro.com.

The views expressed in this article are that of the authors and Wipro does not subscribe to the substance, veracity or truthfulness of the stated opinions.